

1. How many integers between 1 and 60000, inclusive, are divisible by 3 or 5 or 7?

$$n(3|n) = 60000/3 = 20000$$

$$n(5|n) = 60000/5 = 12000$$

$$n(7|n) = 60000/7 = \text{about } 8571$$

$$n(n|3 \cap 5) = 60000/(3 \times 5 = 15) = 4000$$

$$n(n|5 \cap 7) = 60000/(5 \times 7 = 35) = \text{about } 1714$$

$$n(n|3 \cap 7) = 60000/(3 \times 7 = 21) = \text{about } 2857$$

$$n(n|3 \cap 5 \cap 7) = 60000/(3 \times 5 \times 7 = 105) = \text{about } 571$$

$$n(n|3 \cup 5 \cup 7) = n(3|n) + n(5|n) + n(7|n) - n(n|3 \cap 5) - n(n|5 \cap 7) - n(n|3 \cap 7) + n(n|3 \cap 5 \cap 7)$$

$$n(n|3 \cup 5 \cup 7) = 20000 + 12000 + 8571 - 4000 - 1714.28 - 2857.14 + 571.42 = \mathbf{32571}$$

Thus through adding how many integers are divisible by each single number before subtracting the duo-divisible integers and adding the trio-divisible integers, the resulting amount of integers divisible by 3, 5, or 7 is **32571 integers**.

2. An ISU Computer Science shirt is sold in 7 colors, 5 sizes, striped or solid, and long sleeve or short sleeve. (a) How many different shirts are being sold? (b) What if the red and gold shirts do not come striped?

2A.

$$(7 \text{ colors})(5 \text{ sizes})(\text{Striped or solid} \rightarrow 2)(\text{Long or short sleeve} \rightarrow 2)$$

$$(7)(5)(2)(2) = 140$$

140 different shirts are being sold.

2B.

Since only two colors do not come striped, the rest of the colors are still normal. Therefore,

$$(5)(5)(2)(2) = 100 \text{ different shirts from 5 normal colors.}$$

Now regarding the two colors that only come in solid, the calculation is easily adjusted by switching the 2 representing striped or solid to 1, while only inserting 2 for the color input.

$$(2)(5)(1)(2) = 20 \text{ different shirts are being sold from 2 adjusted colors.}$$

Therefore the total amount of shirts when red and gold shirts are not striped is $100 + 20 = 120$ different shirts are being sold.

3. Let A and B be sets of 6 elements and 8 elements, respectively. (a) How many different functions possible from A to B? from B to A? (b) How many different relations possible from A to B? (c) How many of the functions from A to B are one-to-one? (d) How many of the functions from B to A are onto?

3A.

$$A = 6 \text{ elements. } B = 8 \text{ elements. } |A| = 6 \quad |B| = 8$$

$A \rightarrow B$ # of functions = $|B|^{|A|} = 8^6 = \mathbf{262144 \text{ functions}}$

$B \rightarrow A$ # of functions = $|A|^{|B|} = \mathbf{1679616 \text{ functions}}$

3B.

$A \rightarrow B$ # of relations = $2^{(A \times B)} = 2^{(6 \times 8)} = 2^{48} = \mathbf{281474976710656 \text{ relations.}}$

3C.

Calculation is $n!/(n-m)!$ where $m \leq n$.

$A \rightarrow B$ # of one-one functions = $8!/(8-6)! \rightarrow 8!/2! = 40320/2 = 20160$

20160 functions from A to B are one to one

3D.

COMBINATION: $C(x, y) = x! / (y! \times (x - y)!)$

Evens are negative, odds are positive.

1. $6^8 \times C(6, 0) = 6^8 = 1679616$
2. $(C(6, 1)) \times (6-1)^8 = -6 \times 5^8 = -2343750$
3. $(C(6, 2)) \times (6-2)^8 = 15 \times 4^8 = 983040$
4. $(C(6, 3)) \times (6-3)^8 = -20 \times 3^8 = -131220$
5. $(C(6, 4)) \times (6-4)^8 = 15 \times 2^8 = 3840$
6. $(C(6, 5)) \times (6-5)^8 = -6 \times 1^8 = -6$
7. $(C(6, 6)) \times (6-6)^8 = 0$

$1679616 - 2343750 + 983040 - 131220 + 3840 - 6 + 0 = \mathbf{191520 \text{ onto functions from A to B.}}$

4. A sack contains 40 movie tickets, 5 for each of 8 different movies. Five friends want to go to a movie. How many tickets would you have to remove from the sack to guarantee that everyone will be able to watch the same movie? What principle did you use? What if everyone wants to go to 'Oppenheimer'? How many tickets would you have to remove from the sack in that case?

Through the pigeonhole principle stating that if there are n holes and $n + 1$ pigeons, then there is one hole that will contain more than one pigeon, this problem can be effectively solved.

For the first part of the problem, the worst case scenario must be taken into account.

First 8 tickets pulled from the sack will be of the 8 different movies.

This is then repeated three more times until there are 4 tickets of each different movie pulled out of the sack, meaning that $4 \times 8 = 32$ tickets have been pulled out of the sack.

Finally, the next ticket pulled out of the sack must be the fifth ticket of a movie, therefore one would have to remove 33 tickets to have five tickets of the same movie.

And for all of the friends to go to the same specific movie, **all 40 tickets must be removed from the sack**, as this is the worst case scenario due to the principle explained above. Of course, the friends may get lucky and pull the first five tickets of the specific movie, but only the worst case scenario guarantees it.

5. In how many ways can a photographer arrange 8 people in a row from a family of 12 people, if (a) Mom and Dad are in the photo, (b) Mom and Dad are next to each other in the center of the photo (three people on each side of them).

Using combination equation $C(x, y) = \frac{x!}{y! X (x - y)!}$ this problem can be solved.

5A.

Firstly, since Mom and Dad are already in photo, only 6 more people need to be chosen, while there are only 10 people left to choose from.

$$C(10, 6) = \frac{10!}{6! \times (10 - 6)!} = \frac{10!}{6! \times 4!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1} = \frac{5040}{24} = 210$$

Then you must account for actually arranging the people in the photo.

You can do this by using 8! to represent the different arrangements.

$210 \times 8! = 8467200$ different ways to arrange the people.

5B.

Since the mom and the dad are in the photo, still need to choose 6 more people to be in picture.

This is portrayed by $C(10, 6)$, which as found above, equals 210.

Mom and Dad = One unit if always together. Therefore,

7! ways to arrange the people in the photo.

Finally, the mom and the dad can be arranged in two different ways, so their arrangement possibilities = 2.

Multiply the three numbers found above,

$210 \times 7! \rightarrow 5040 \times 2 = 2116800$ different ways to arrange people for the photo.

6. A coin is flipped nine times where each flip comes up either head or tails. How many possible outcomes contain at least five heads?

Using a sum of combination equations $C(x, y) = \frac{x!}{y! X (x - y)!}$, this problem can be solved.

$$\text{Outcomes with exactly 5 heads} = C(9, 5) = \frac{9!}{5! X (9 - 5)!} \rightarrow \frac{9!}{5! X 4!} \rightarrow \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1}$$

$$\frac{3024}{24} = 126 \text{ outcomes}$$

$$\text{Outcomes with exactly 6 heads} = C(9, 6) = \frac{9!}{6! X (9 - 6)!} \rightarrow \frac{9!}{6! X 3!} \rightarrow \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 3 \times 2 \times 1}$$

$$\frac{504}{6} = 84 \text{ outcomes}$$

$$\text{Outcomes with exactly 7 heads} = C(9, 7) = \frac{9!}{7! X (9 - 7)!} \rightarrow \frac{9!}{7! X 2!} \rightarrow \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 2 \times 1}$$

$$\frac{72}{2} = 36 \text{ outcomes}$$

Outcomes with exactly 8 heads = $C(9, 8) = 9! / 8! \times (9 - 8)! \rightarrow 9! / 8! \times 1! \rightarrow 9 / 1$
 $9/1 = 9$ outcomes

Outcomes with exactly 9 heads = $C(9, 9) = 9! / 9! \times (9 - 9)! \rightarrow 9! / 6! \times 0! \rightarrow 1 / 1$
 $= 1$ outcome

$126 + 84 + 36 + 9 + 1 = 256$ outcomes with at least 5 heads.

7. 12 women and 10 men are on the faculty. How many ways are there to pick a committee of 6 if (a) Ann and Beth will not serve together, (b) at least one woman must be chosen, (c) at least one man and one woman must be chosen.

Total amount of possible committee combinations:

$12 + 10 = 22$ people and 6 positions on the committee $\rightarrow C(22, 6) = 22! / 6! \times (22 - 6)!$
 $22!/6! \times 16! \rightarrow 22 \times 21 \times 20 \times 19 \times 18 \times 17 / 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 74613$ total possible combinations.

7A. Easiest way to find amount where they don't work together is to find amount where they do and subtract total by number.

$6 - 2 = 4$ positions left. $22 - 2 = 20$ people left. $\rightarrow C(20, 4) = 20! / 4! \times (20 - 4)! \rightarrow$
 $20 \times 19 \times 18 \times 17 / 4 \times 3 \times 2 \times 1 \rightarrow 116280 / 24 = 4845$ combinations

$74613 - 4845 = 69768$ combinations where Ann and Beth do not serve together.

7B. Easiest way to solve this is to find all male committee combination amount and subtract from total amount.

10 males and 6 spots on committee, so $C(10, 6) = 10! / 6! \times (10 - 6)! = 10! / 6! \times 4! =$
 $10 \times 9 \times 8 \times 7 / 4 \times 3 \times 2 \times 1 \rightarrow 5040 / 24 = 210$ combinations of all-men committees.

$74613 - 210 = 74403$ combinations that have at least one woman.

7C. Easiest way to solve is to also find the all women committee combinations and subtract that along with the all men combinations from the total.

12 females and 6 spots on committee, so $C(12, 6) = 12! / 6! \times (12 - 6)! = 12! / 6! \times 6! =$
 $12 \times 11 \times 10 \times 9 \times 8 \times 7 / 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 665280 / 720 = 924$ combinations of all-women committees.

924 all women committees + 210 all men committees = 1134

$74613 - 1134 = 73479$ committees with at least one man and woman.

8. Prove, using a combinatorial argument, that $C(m + n, 2) = C(m, 2) + C(n, 2) + mn$,
where $m, n \geq 2$

Here is original combination formula.

$C(m, n) = m! / n! \times (m - n)!$

Plugging in $m + n$ for m .

$C(m + n, 2) = (m + n)! / 2! \times (m + n - 2)!$

Rewriting $(m+n)!$ In order to eliminate $(m+n-2)!$ From numerator/denominator.

$$(m+n)(m+n-1)\cancel{(m+n-2)!} / 2! \times \cancel{(m+n-2)!} \text{ And } 2! = 2$$

Rewrite and factor out.

$$(m+n)(m+n-1) / 2 = (m^2 + n^2 - n - m + 2m^n) / 2$$

Separate by like terms.

$$(m^2 - m) / 2 + (n^2 - n) / 2 + 2mn / 2$$

$$m(m-1)/2 + n(n-1)/2 + mn$$

Multiply both sides of fraction by $m/n - 2$ factorial to fit equation.

$$m(m-1)(m-2)!/2! \times (m-2)! + n(n-1)(n-2)!/2!(n-2)! + mn$$

Rewrite factorials.

$$m!/2! \times (m-2)! + n!/2! \times (n-2)! + mn$$

You can substitute combination equations now because the fractions now fit the combination between the variable and the 2.

$$C(m, 2) + C(n, 2) + mn$$

$$\textbf{Therefore, } C(m+n, 2) = C(m, 2) + C(n, 2) + mn$$

9. Prove, (a) using a combinatorial argument, and (b) using an algebraic proof, that $P(n, 3)C(n-3, k-3) = C(n, k)P(k, 3)$.

9A.

Assume $n = 6$ and $k = 4$, so that...

$$P(n, 3) = P(6, 3) = 6!/(6-3)! = 6!/3!$$

$$6!/3! = 6 \times 5 \times 4 = 120$$

$$C(n-3, k-3) = C(3, 1) = 3!/1! \times (3-1)! = 3!/2!$$

$$3!/2! = 3$$

$$\text{LHS: } 120 \times 3 = 360$$

$$C(n, k) = C(6, 4) = 6!/4! \times (6-4)! = 6!/4! \times 2!$$

$$6!/4! \times 2! = 6 \times 5 / 2 = 15$$

$$P(4, 3) = 4!/(4-3)! = 4!/1!$$

$$4!/1! = 4 \times 3 \times 2 = 24$$

$$\text{RHS: } 24 \times 15 = 360$$

360 = 360, therefore the statement is proved through a combinatorial argument.

9B.

$$P(n, 3) = n!/(n-3)!$$

$$C(n-3, k-3) = (n-3)!/(k-3)!(n-k)!$$

$$C(n, k) = n!/k!(n-k)!$$

$$P(k, 3) = k!/(k-3)!$$

$$\text{LHS: } n!/\cancel{(n-3)!} \times \cancel{(n-3)!}/(k-3)!(n-k)!$$

$$\rightarrow n!/(k-3)!(n-k)!$$

RHS: $n!/\cancel{k!}(n-k)! \times \cancel{k!}/(k-3)!$

$\rightarrow n!/(n-k)! \times (k-3)!$

LHS = RHS: $n!/(k-3)!(n-k)! = n!/(n-k)!(k-3)!$ 

After simplifying both sides of the equation, it is clear that the statement holds true algebraically.

Expanding the combination and permutation equations into terms/variables allows us to this conclusion.